

AWS IAM Managed Policies

To Begin with the Lab

Summary of the Lab

In this lab, two IAM users were created to understand how **AWS Managed Policies** provide permissions to AWS services. Initially, both users were created without any permissions and were unable to access AWS resources. The **AmazonEC2FullAccess** managed policy was then attached to the first user, allowing full management of Amazon EC2 resources along with permissions to related services such as Amazon VPC and Amazon CloudWatch. The second user received the **AmazonVPCFullAccess** managed policy, demonstrating that the user could manage VPC resources but could not launch EC2 instances. Finally, the lab highlighted both the advantages and limitations of AWS Managed Policies and explained why Customer Managed Policies are needed for more granular permission control.

Understanding AWS IAM Managed Policies

An **AWS Managed Policy** is a predefined IAM policy created and maintained by AWS. It exists to quickly grant permissions for common AWS job roles without requiring administrators to write JSON policy documents manually. Since AWS maintains these policies, they are automatically updated whenever AWS introduces new permissions or related services. However, because they are generic, they often grant broader permissions than required, making them less suitable when fine-grained access control is needed.

Example:

An employee responsible for managing Amazon EC2 instances can be assigned the **AmazonEC2FullAccess** managed policy. This allows them to launch, stop, terminate, and manage EC2 instances, while also automatically providing access to related services such as Amazon VPC networking and Amazon CloudWatch monitoring.

Lab Steps

Step 1 – Create the First IAM User

- Sign in to the AWS Management Console using the root user.
- Navigate to **IAM** → **IAM Users**.

Identity and Access Management (IAM)

Search IAM

Dashboard

▼ Access Management

- Roles
- Policies
- IAM users**
- IAM user groups
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

▼ Access reports

- Access Analyzer
- Resource analysis
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies

IAM Dashboard Info

Security recommendations 1

- Root user has MFA
Having multi-factor authentication (MFA) for the root user improves security for this account.
- Add MFA for yourself
Add multi-factor authentication (MFA) for yourself to improve security for this account.
- Your user, cf-shashank1, does not have any active access keys that have been unused for more than a year.
Deactivating or deleting unused access keys improves security.

IAM resources Info

Resources in this AWS Account

You are approaching your quota of available IAM resources. [Learn more](#)

User groups	Users	Roles	Policies	Identity providers
42	8	415	1132 368 available	1

What's new Info

Updates for features in IAM

AWS Account

Account ID
[Redacted]

Account Alias
[Create](#)

Sign-in URL for IAM users in this account
[https://\[Redacted\].signin.aws.amazon.com/console](https://[Redacted].signin.aws.amazon.com/console)

Quick Links

[My security credentials](#)

Manage your access keys, multi-factor authentication (MFA) and other credentials.

Tools Info

[Policy simulator](#)

The simulator evaluates the policies that you choose and determines the effective permissions for each of the actions that you specify.

- Click **Create user**.

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Managing human user access account by account? There's a better way.

Streamline human access to AWS and cloud apps when you enable Identity Center. [Learn more](#)

[Watch how it works](#)

One-time set up for workforce user access

Centrally manage access to multiple AWS accounts

Provide access centrally to the cloud applications your workforce uses

All with one-click access through a simple web portal

IAM users (6) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Access key ID
☐	[Redacted]	/	0	10 hours ago	-	6 days	16 hours ago	Active - [Redacted]
☐	[Redacted]	/	0	1 hour ago	-	16 days	1 hour ago	-
☐	[Redacted]	/	0	21 hours ago	-	5 days	21 hours ago	-
☐	[Redacted]	/	1	5 days ago	-	5 days	5 days ago	-
☐	[Redacted]	/	1	18 hours ago	-	19 hours	19 hours ago	Active - [Redacted]
☐	[Redacted]	/	1	10 hours ago	-	10 hours	-	-

[Delete](#) [Create user](#)

- Enter the username **EC2User**.
- Enable **Provide user access to the AWS Management Console**.
- Create a custom password.
- Disable **User must create a new password at next sign-in** only for this lab purpose.
- Click **Next**.

User details

User name
EC2User

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , @ _ - (hyphen)

☒ **Provide user access to the AWS Management Console** *optional*
In addition to console access, users with programmatic access can use the same console credentials for programmatic access without the need for access keys.

Are you providing console access to a person?

User type

☐ Specify a user in Identity Center - Recommended
We recommend that you use Identity Center to provide console access to a person. With Identity Center, you can centrally manage user access to AWS IAM resources and applications.

☒ **I want to create an IAM user**
We recommend that you create IAM users only if you need to enable programmatic access through access keys, service-specific credentials for AWS CodeCommit or Amazon Keyspaces, or a backup credential for emergency account access.

Console password

☐ Autogenerated password
You can view the password after you create the user.

☒ **Custom password**
Specify a password for the user.

Must be at least 8 characters long
Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = [] { } | ' "

☐ Show password

☒ **Users must create a new password at next sign-in - Recommended**
Users automatically get the [IAMUserChangerPassword](#) policy to allow them to change their own password.

- Do not assign any permissions.
- Click **Create user**.

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name EC2User	Console password type Custom password	Require password reset No
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Permissions summary

Name	Type	Used as
No resources		

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

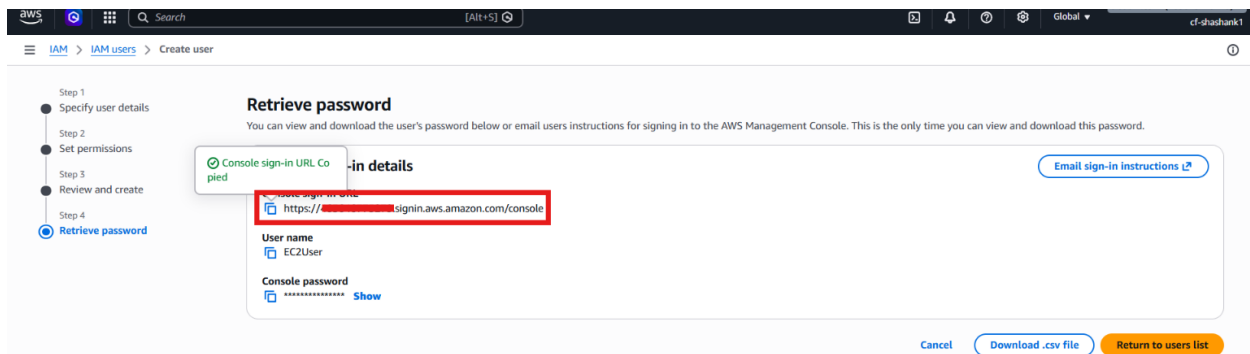
[Add new tag](#)

You can add up to 50 more tags.

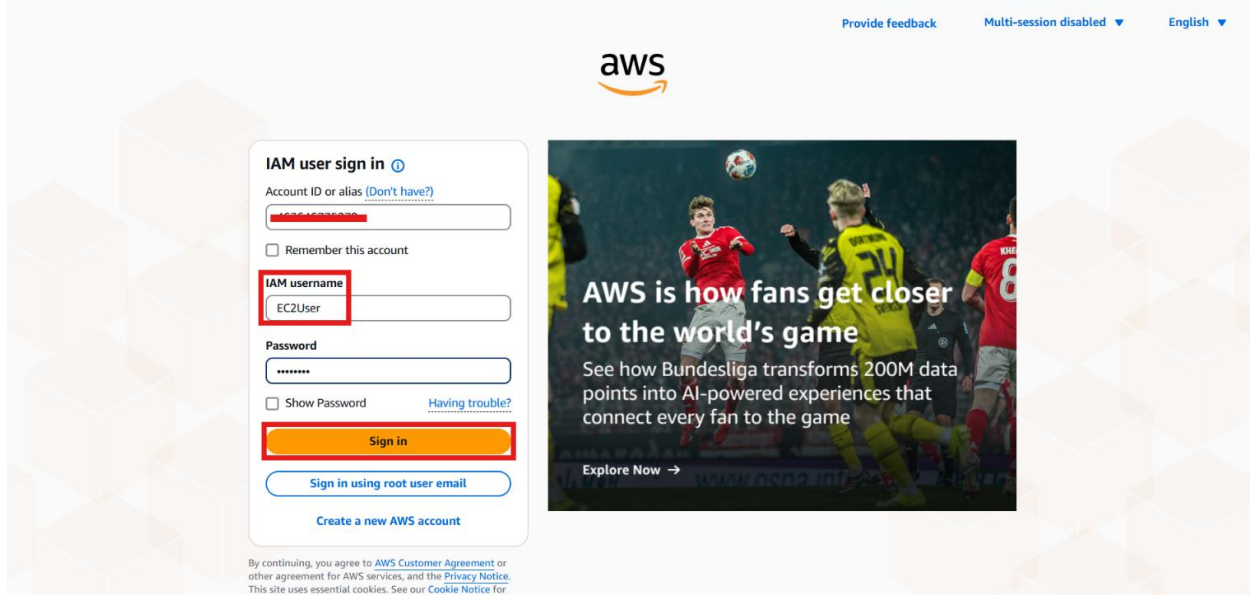
[Cancel](#) [Previous](#) [Create user](#)

Step 2 – Verify No Permissions

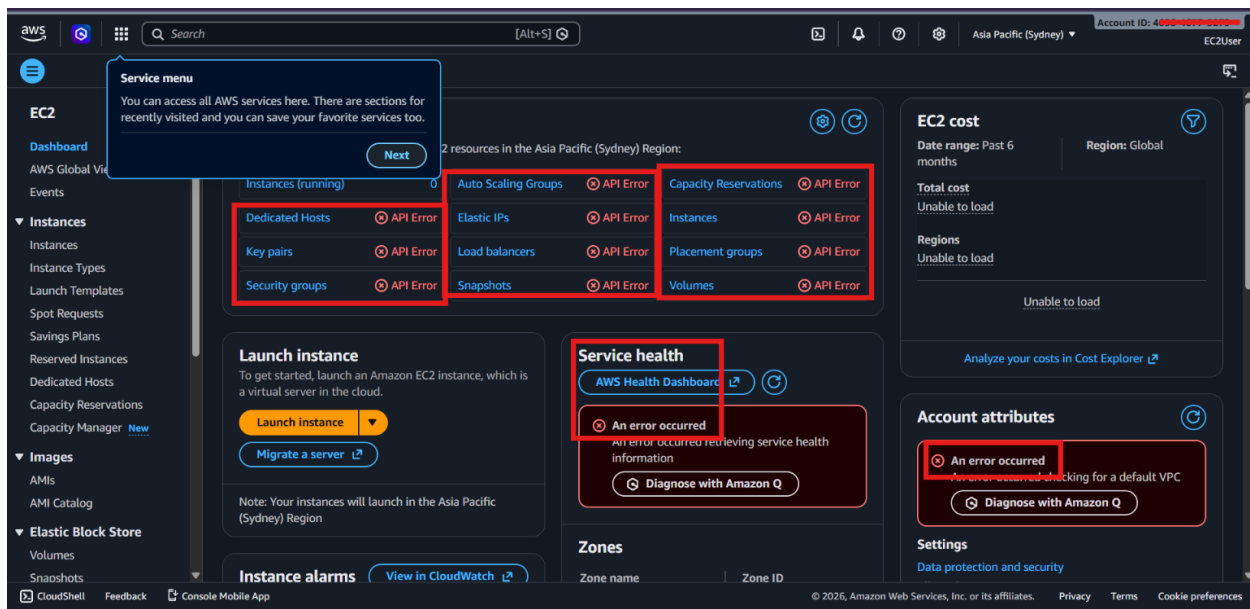
- Copy the IAM user sign-in URL.



- Open the sign-in URL in another browser or Incognito window.
- Log in as **EC2Mastermind**.

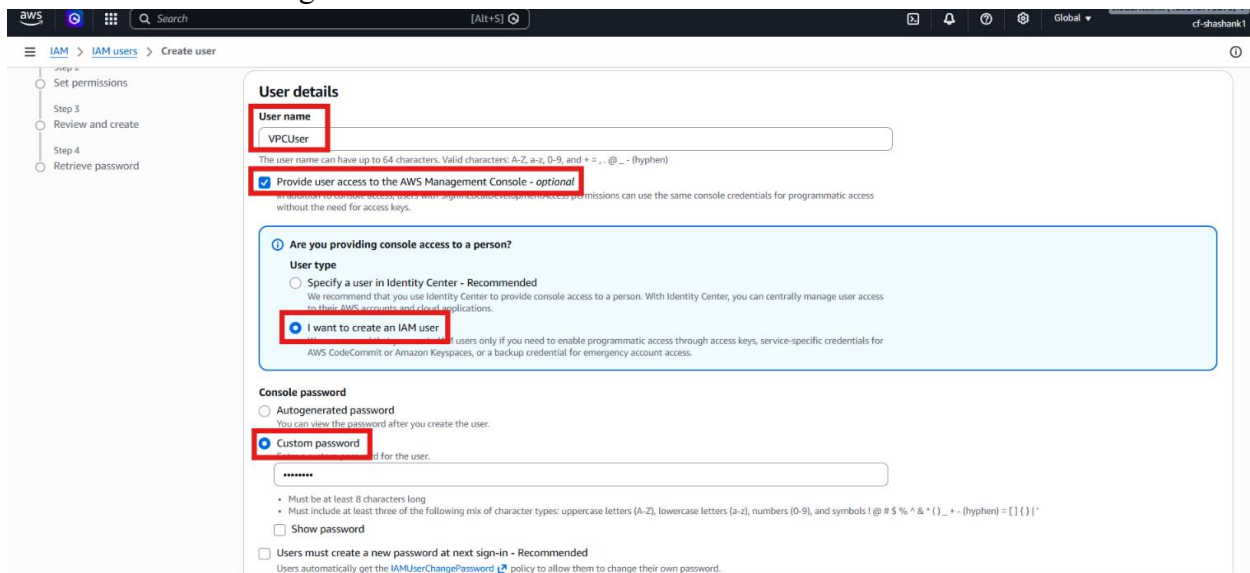


- Open **Amazon EC2**.
- Verify that API authorization errors appear because no permissions are assigned.



Step 3 – Create the Second IAM User

- Return to the root account.
- Navigate to **IAM** → **Users**.
- Click **Create user**.
- Enter the username **VPCUser**.
- Enable AWS Management Console access.



- Create a password or leave it autogenerated.
- Leave the user without permissions.
- Click **Create user**.

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name: VPCUser | Console password type: Custom password | Require password reset: No

Permissions summary

Name	Type	Used as
No resources		

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#) [Previous](#) [Create user](#)

Step 4 – Attach the Amazon EC2 Managed Policy

- Select the **EC2Mastermind** user.
- Open the **Permissions** tab.
- Click **Add permissions**.

EC2User

Summary

ARN: [arn:aws:iam::163614326299:user/EC2User](#) | Console access: Enabled without MFA | Access key 1: [Create access key](#)

Created: June 25, 2026, 13:41 (UTC+05:30) | Last console sign-in: Today

Permissions | Groups | Tags | Security credentials | Last Accessed

Permissions policies (0)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type: All types

Policy name	Type	Attached via
No resources to display		

Permissions boundary (not set)

[Add permissions](#)

[Add permissions](#)

[Create inline policy](#)

- Select **Attach policies directly**.
- Search for **AmazonEC2FullAccess**.
- Select the policy.
- Click **Next**.

Add permissions
Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

- ☐ Add user to group
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- ☐ Copy permissions
Copy all group memberships, attached managed policies, inline policies, and any existing permissions boundaries from an existing user.
- ☒ **Attach policies directly**
Attach policies directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/2639)

Filter by Type: All types | 1 match

Policy name	Type	Attached entities
☒ AmazonEC2FullAccess	AWS managed	6

[Cancel](#) [Next](#)

- Click **Add permissions**.

Review
The following policies will be attached to this user. [Learn more](#)

User details
User name: EC2User

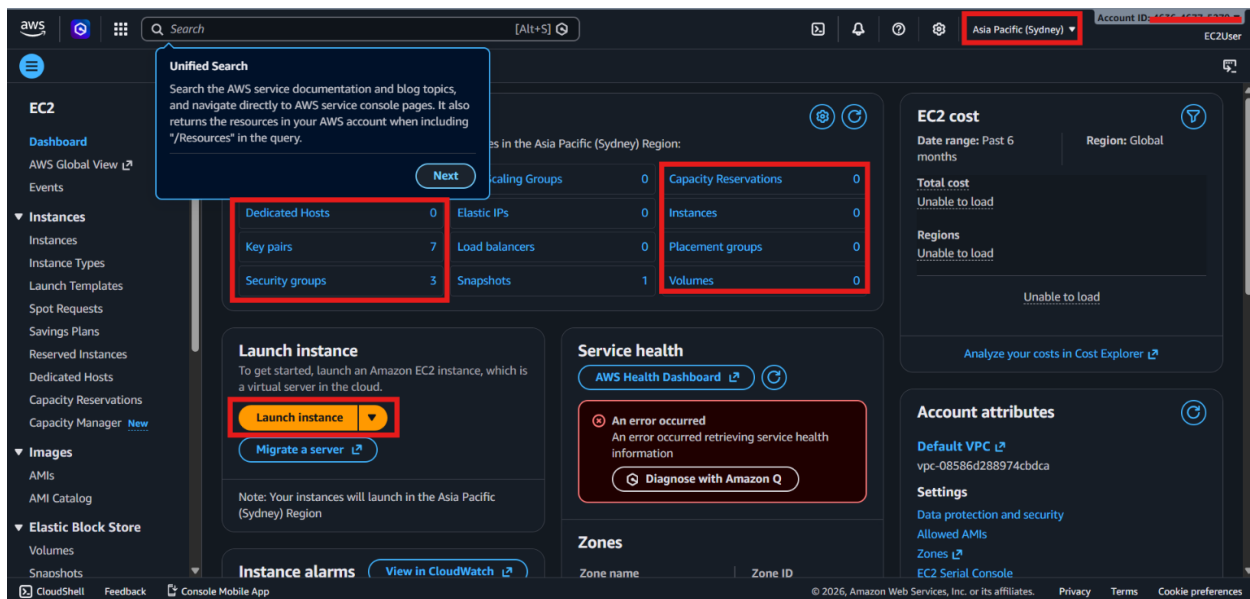
Permissions summary (1)

Name	Type	Used as
AmazonEC2FullAccess	AWS managed	Permissions policy

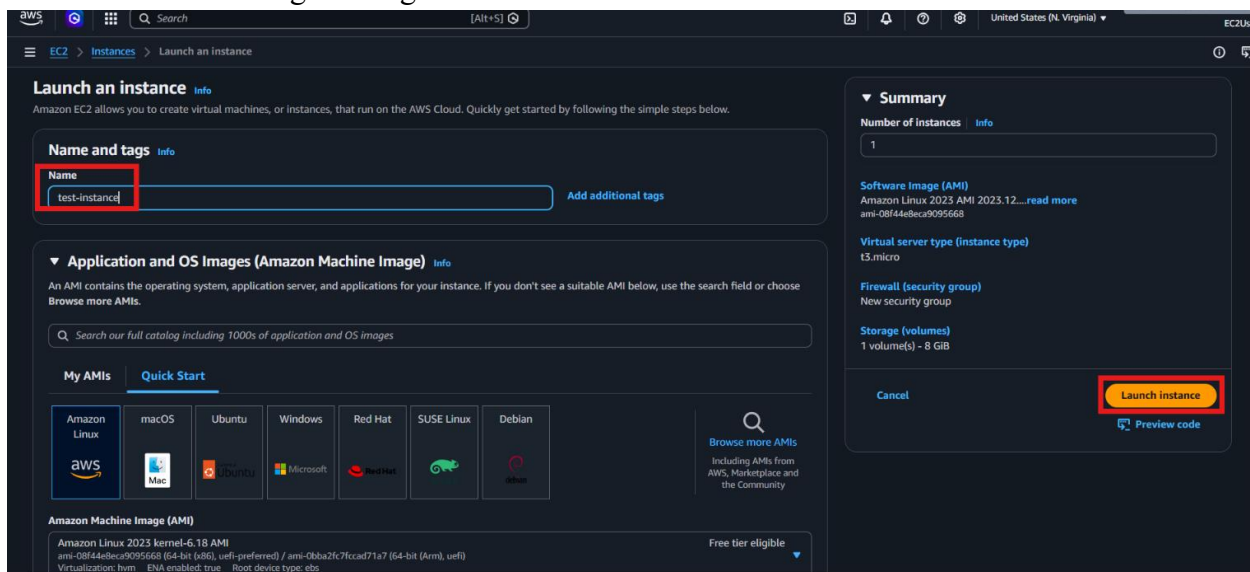
[Cancel](#) [Previous](#) [Add permissions](#)

Step 5 – Verify EC2 Permissions

- Return to the **EC2Mastermind** browser session.
- Refresh the AWS Console.
- Open **Amazon EC2**.
- Verify that EC2 resources are now accessible.



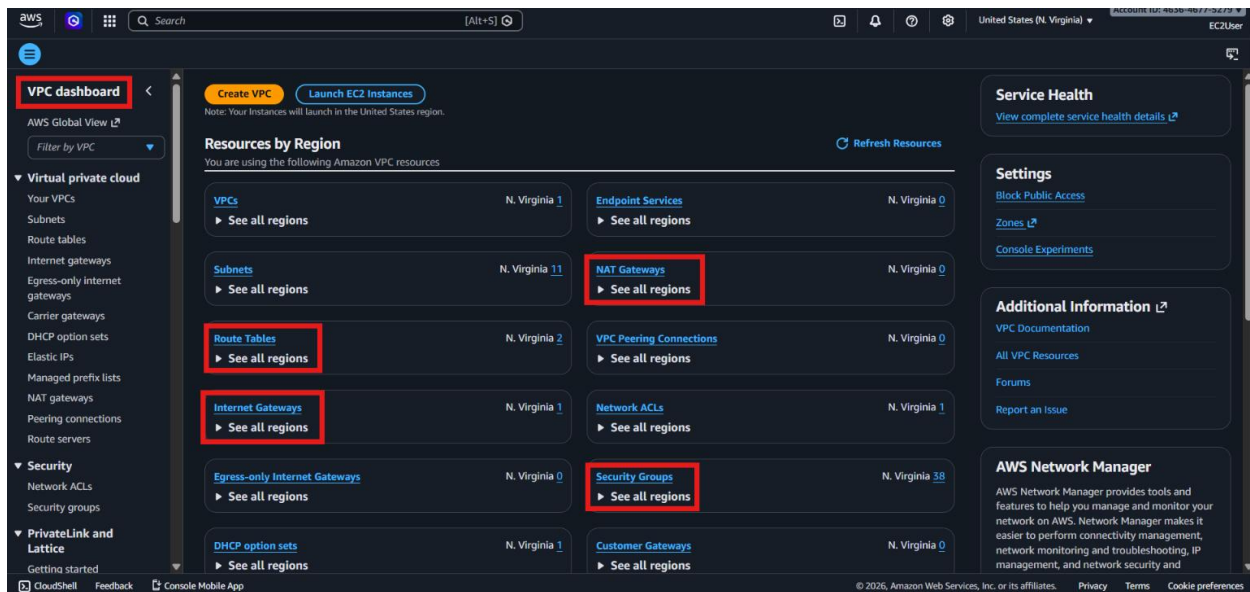
- Launch a test EC2 instance with least instance configuration and key pair created as earlier.
- Make sure to change the region first!.



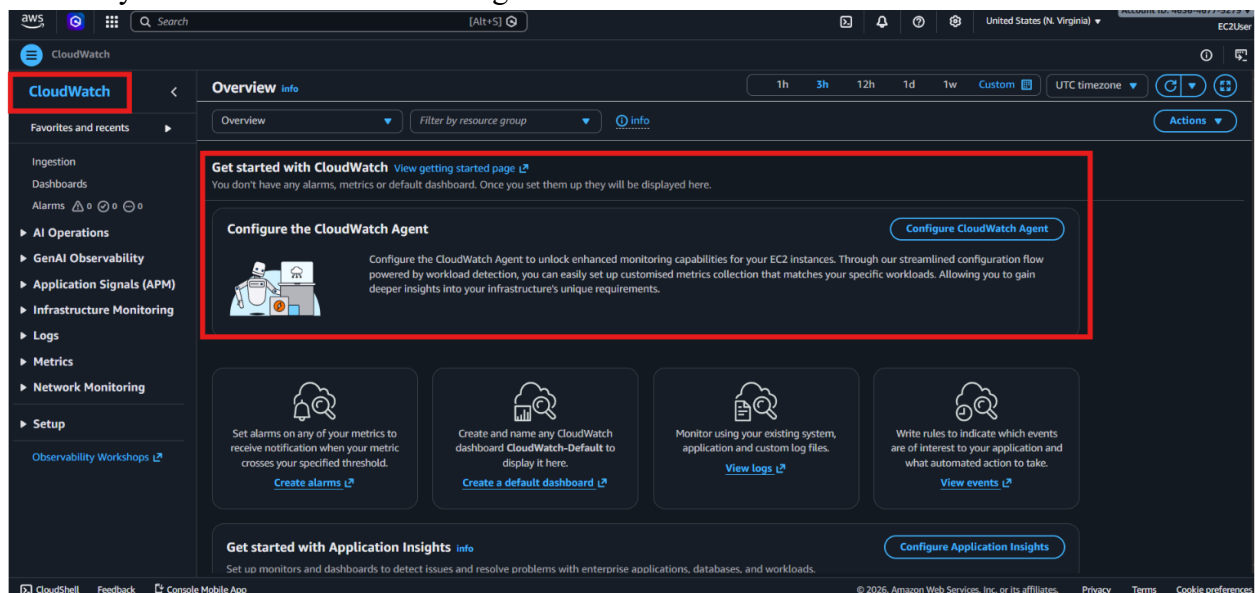
- Verify that the instance launches successfully.

Step 6 – Verify Related Service Permissions

- While logged in as **EC2User**, open **Amazon VPC**.
- Verify that VPC resources such as:
 - Internet Gateways
 - Route Tables
 - Security Groups
 - NAT Gateways
 are accessible.



- Open Amazon CloudWatch.
- Verify that CloudWatch monitoring is also available.

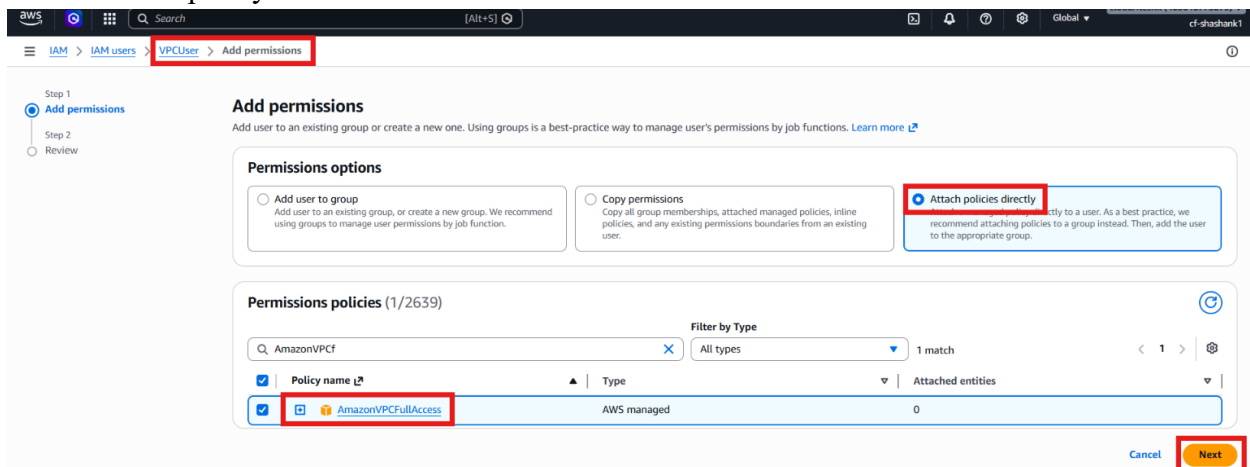


- Observe that these permissions were granted automatically because they are required for EC2 management.

Step 7 – Attach the Amazon VPC Managed Policy

- Return to the root account.
- Select the **VPCUser** user.
- Open **Permissions**.
- Click **Add permissions**.
- Choose **Attach policies directly**.
- Search for **AmazonVPCFullAccess**.

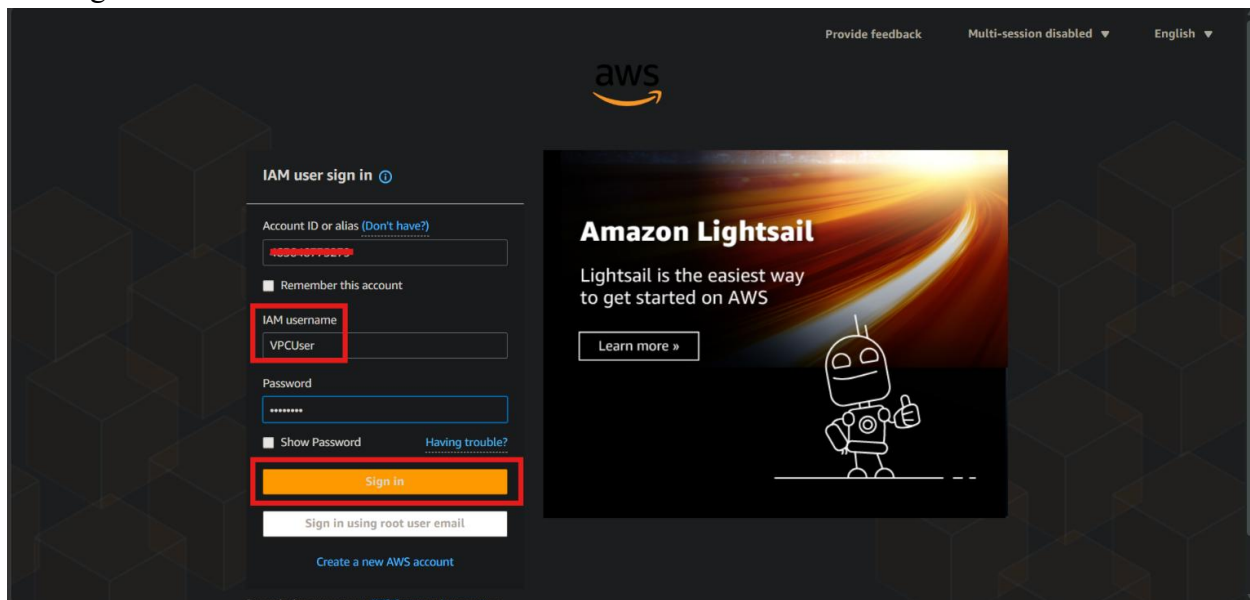
- Select the policy.



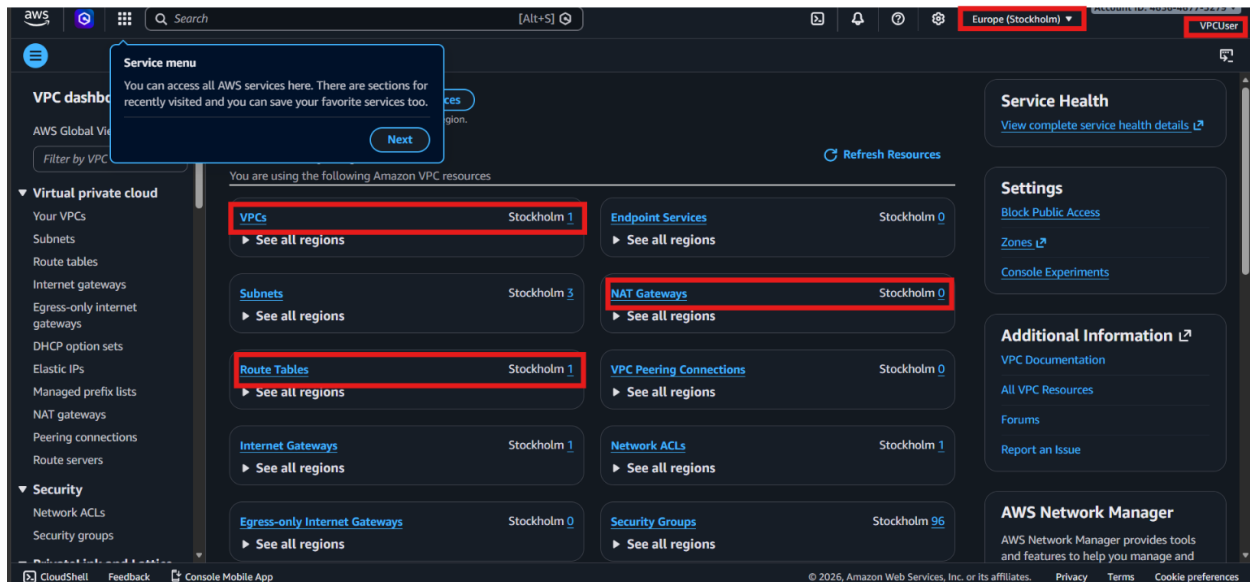
- Click Add permissions.

Step 8 – Verify VPC Permissions

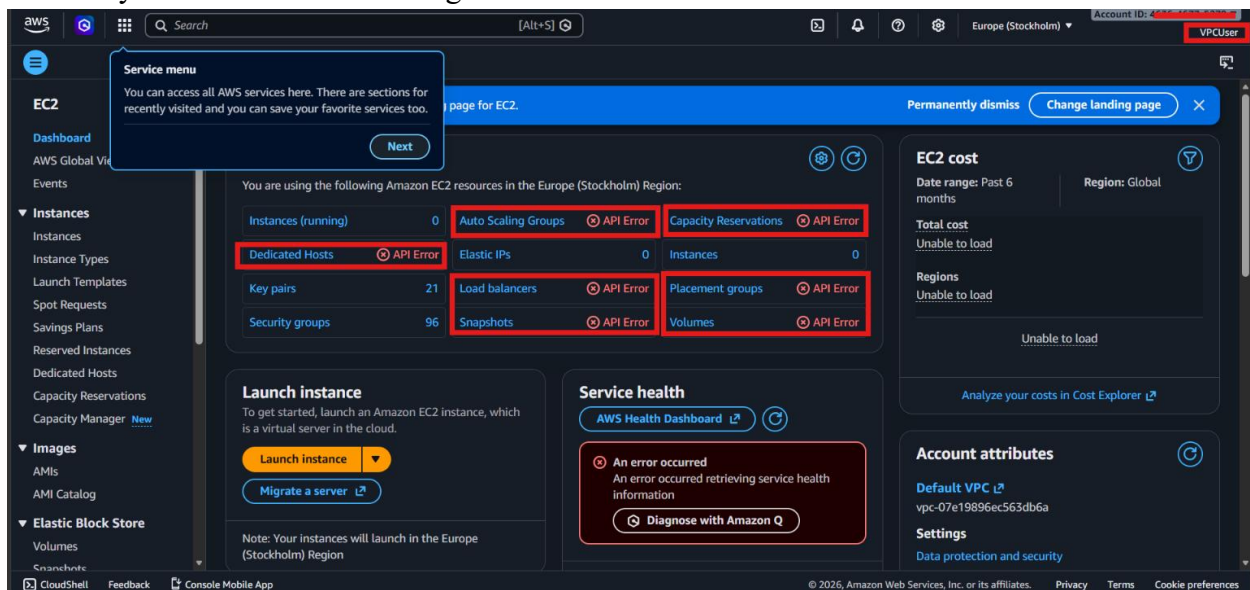
- Log in as VPCUser.



- Open Amazon VPC.
- Verify that VPC resources can be created and managed.



- Make sure to change the **Region** before doing anything to your account.
- Open **Amazon EC2**.
- Verify that EC2 instance management remains restricted.



- Observe that networking-related options are available, but EC2 instance operations are denied.

Step 9 – Compare Managed Policy Behavior

- Compare both IAM users.
- Observe that:
 - **EC2User** can manage EC2 and related networking services.
 - **VPCUser** can manage networking resources but cannot fully manage EC2 instances.
- Understand that AWS Managed Policies automatically include permissions for supporting AWS services.

Verification

- Two IAM users were created successfully.
- Users without policies could not access AWS services.
- **AmazonEC2FullAccess** allowed full EC2 administration.
- The EC2 managed policy also granted access to related services such as **Amazon VPC** and **Amazon CloudWatch**.
- **AmazonVPCFullAccess** allowed VPC administration but restricted EC2 instance management.
- The lab demonstrated the benefits and limitations of AWS Managed Policies and why Customer Managed Policies are used for fine-grained access control.